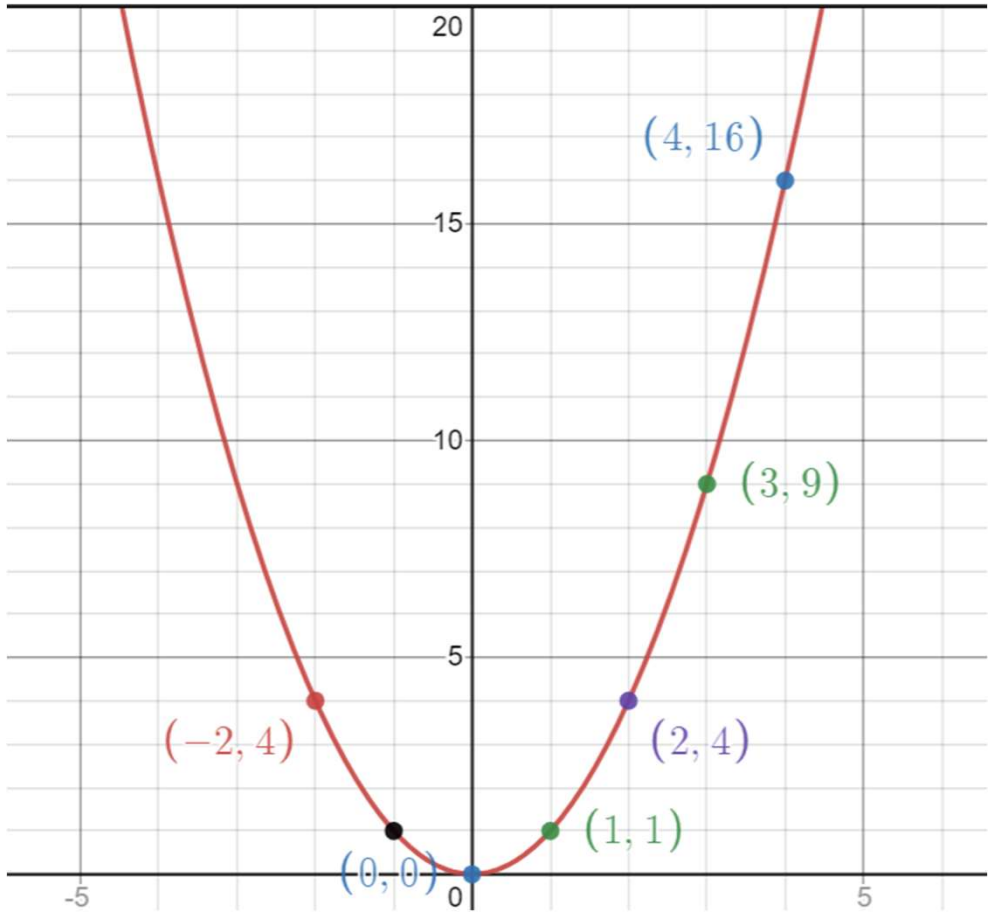
# Quadratic functions and transformations

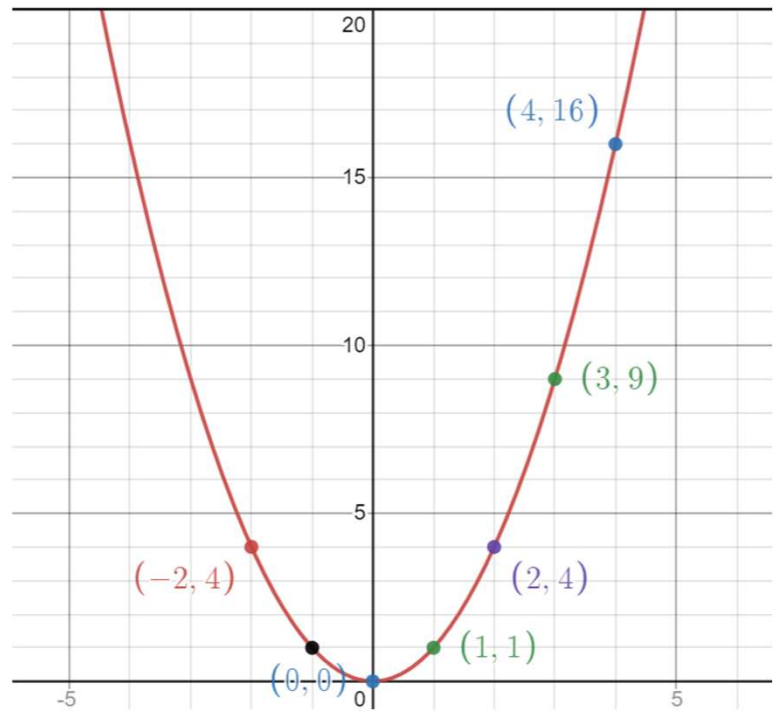
$f(x) = x^2$



We get the parabola shape because of the way the output grows with the input. Because a negative times a negative is a positive, the output is always zero or positive for this function.

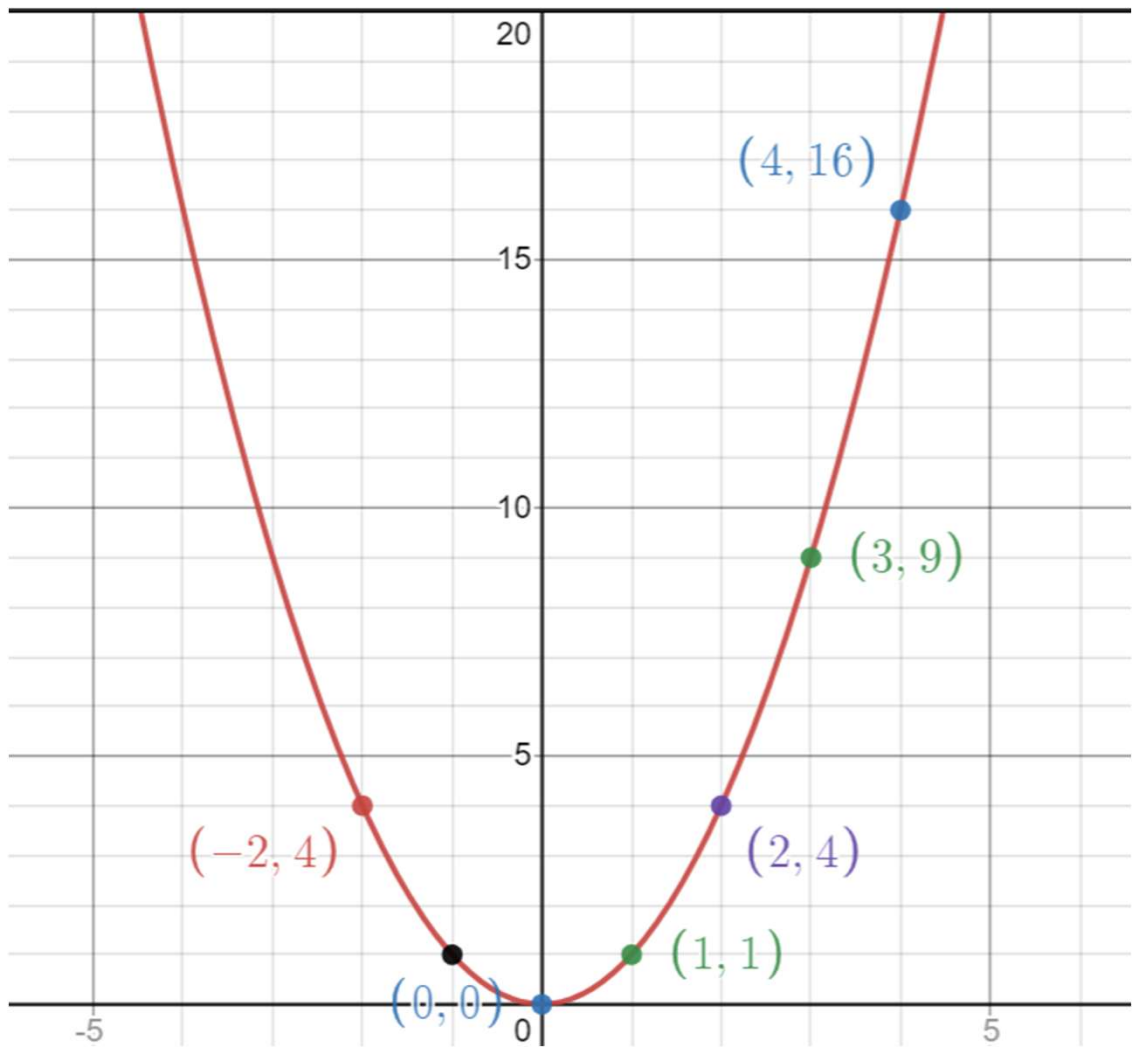
$$f(x) = x^2$$

| $x$ input | $f(x)$ or $y$ output |
|---|---|
| 0 | $0 \cdot 0 = 0$ |
| 1 | $1 \cdot 1 = 1$ |
| 2 | $2 \cdot 2 = 4$ |
| -1 | $(-1)(-1) = 1$ |
| -2 | $(-2)(-2) = 4$ |

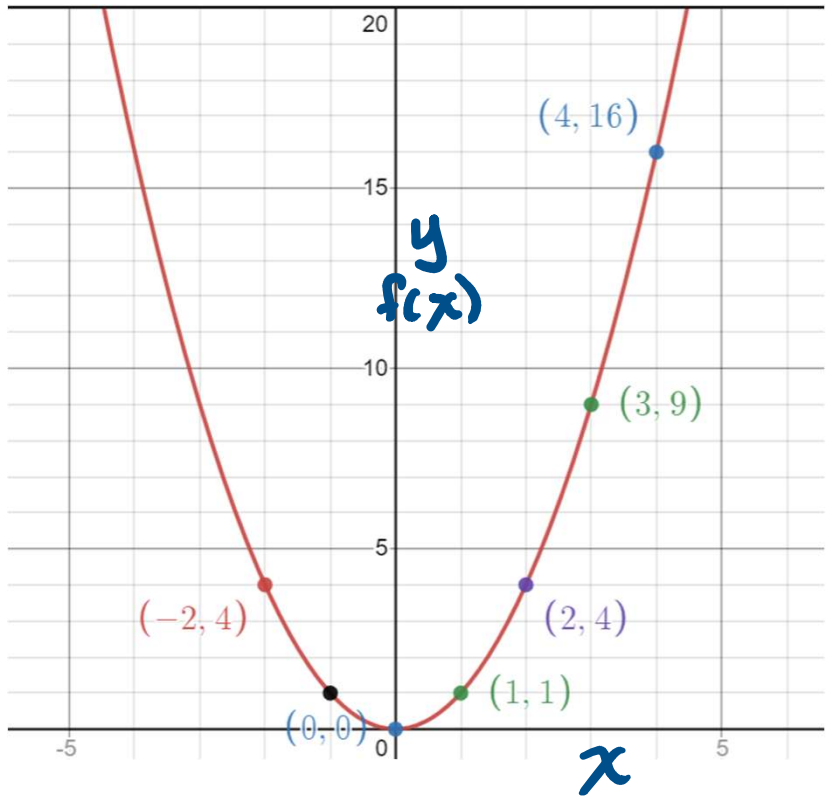


# $\{(x,y): y= x^2\}$

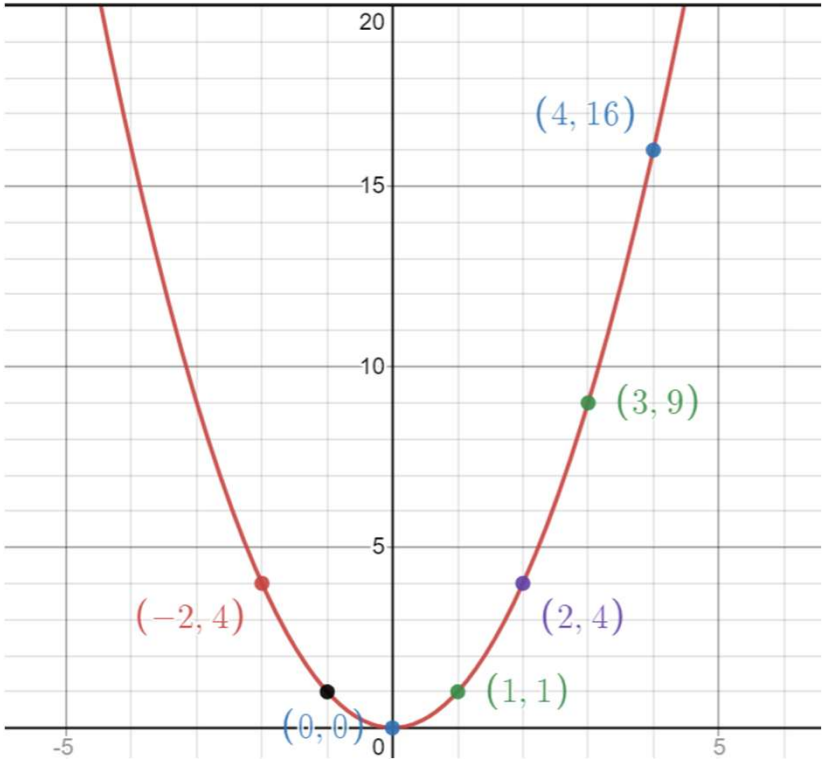
The graph of a quadratic has this shape, called a **parabola**. The point at the bottom is called the **vertex** and is where it changes direction.



Notice that x∈R, and y or f(x) is always positive.



This is called an even function because it reflects or folds across the y axis.

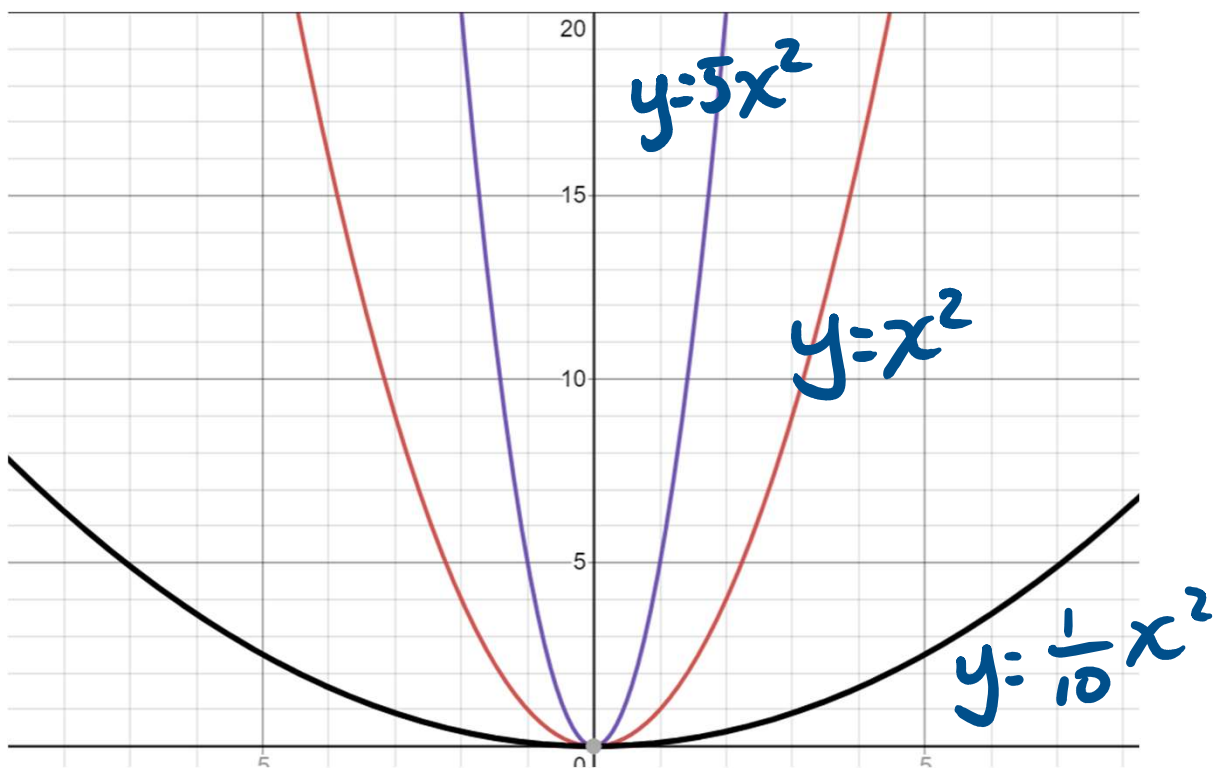


# What happens when we transform our parent function?

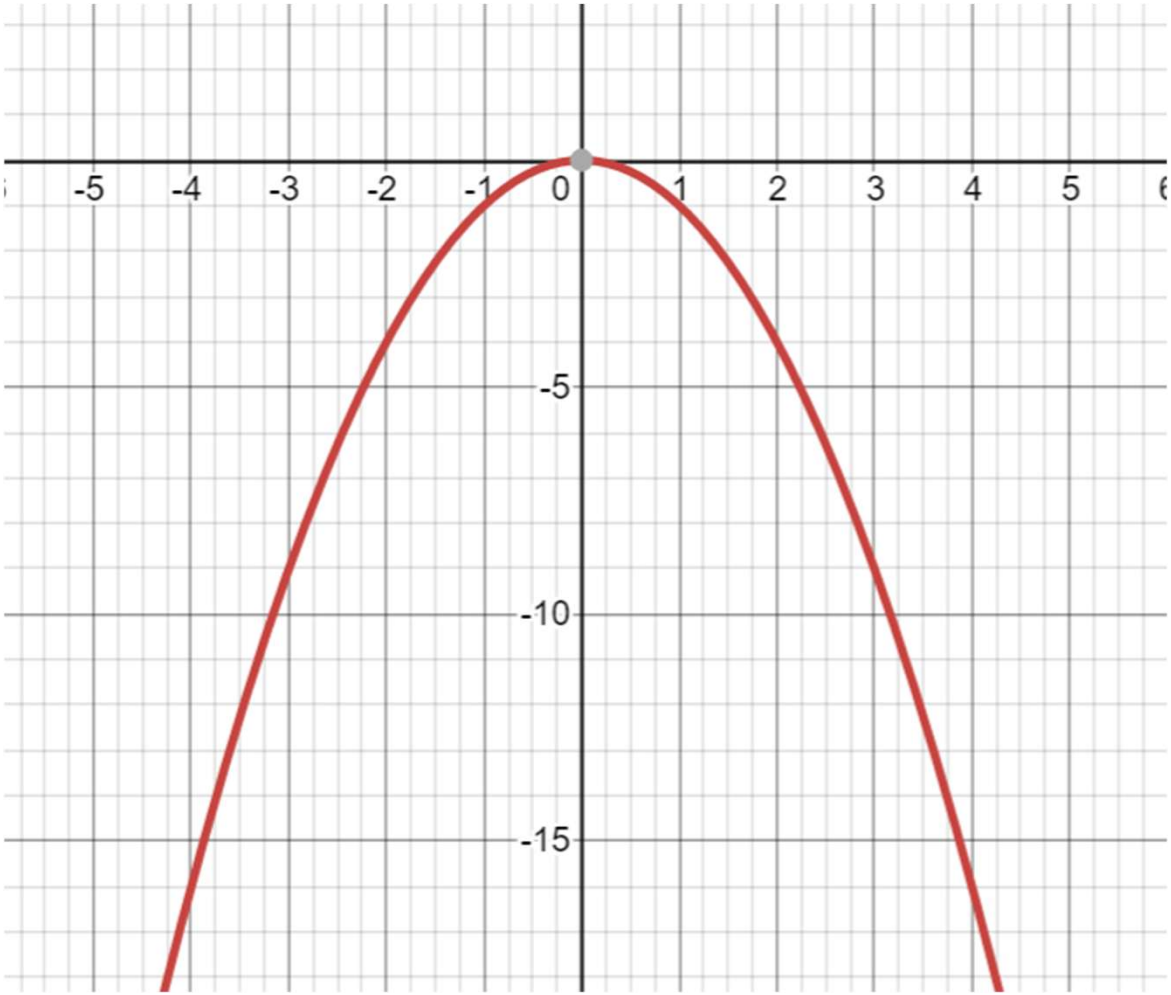
$y = x^2$

$y = 5x^2$

$y = \frac{1}{10}x^2$

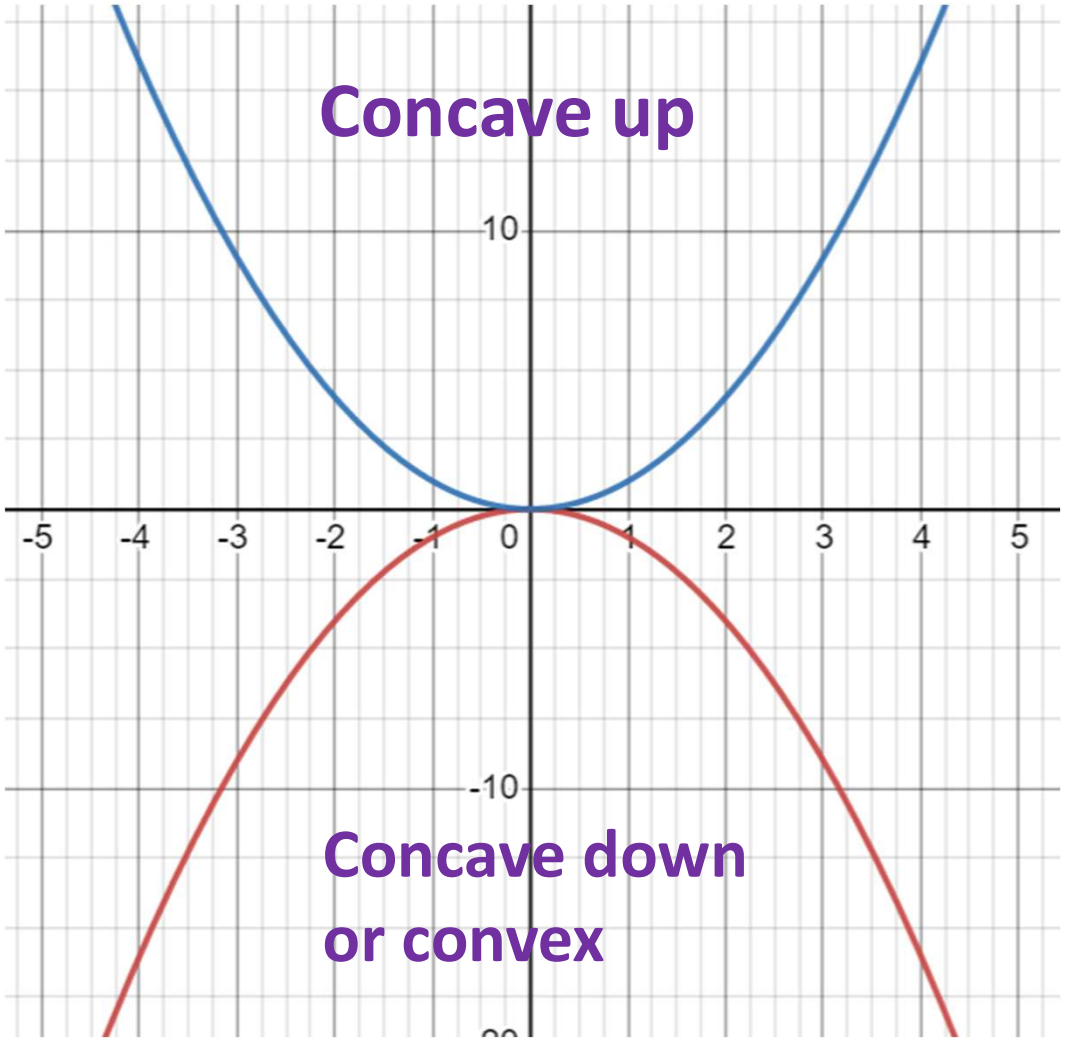


$y=-1x^2$



$y = x^2$

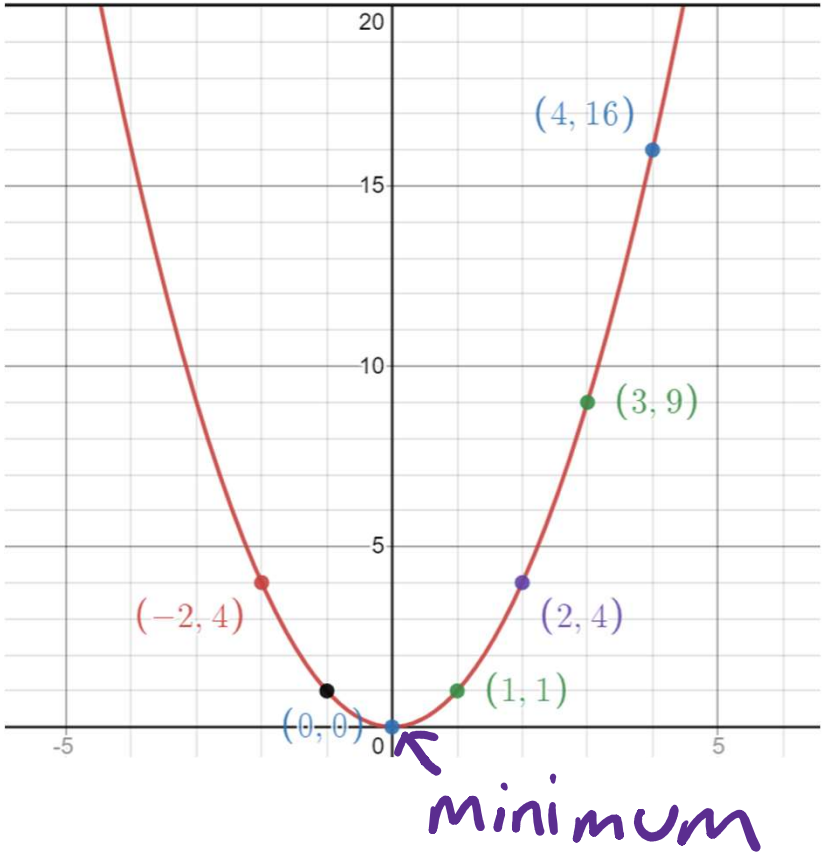
$y = -x^2$



The **vertex** is where the parabola changes direction.

When the leading coefficient is positive, the vertex is the minimum output of the function or just the **minimum** of the function.

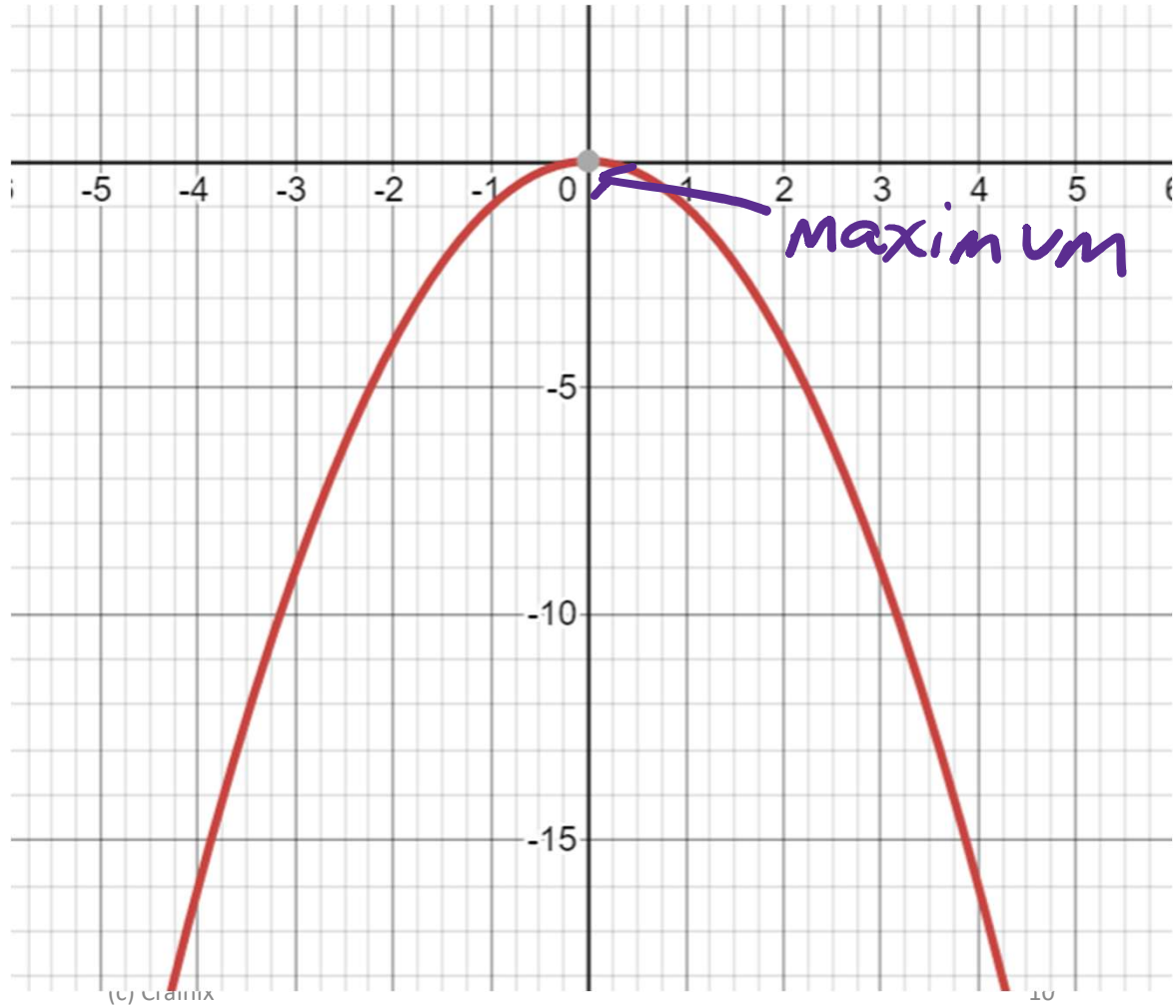
$y = ax^2$



The **vertex** is where the parabola changes direction.

When the leading coefficient is negative, the vertex is the maximum output of the function or the **maximum**.

$y = -ax^2$



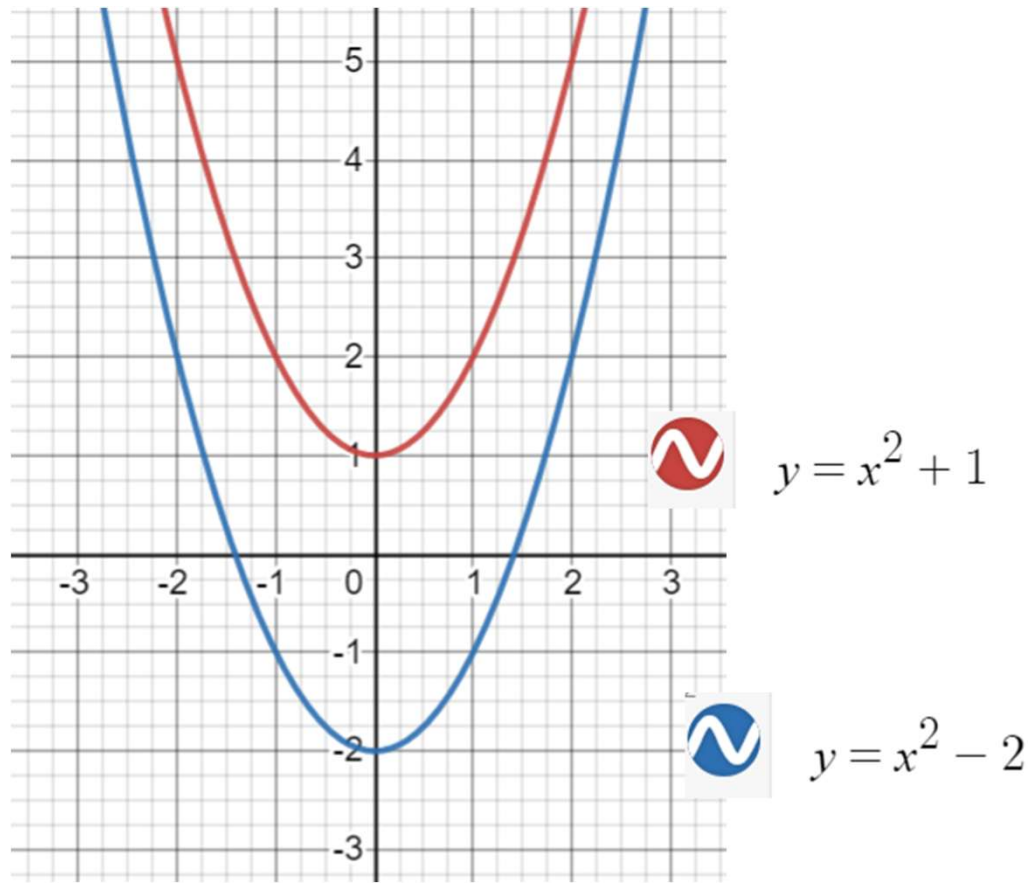
# Adding a positive or negative constant to a quadratic function.

Adding a positive constant shifts the function up by that amount.

$y = x^2 + 1$

Adding a negative constant shifts the function down by that amount.

$y = x^2 - 2$



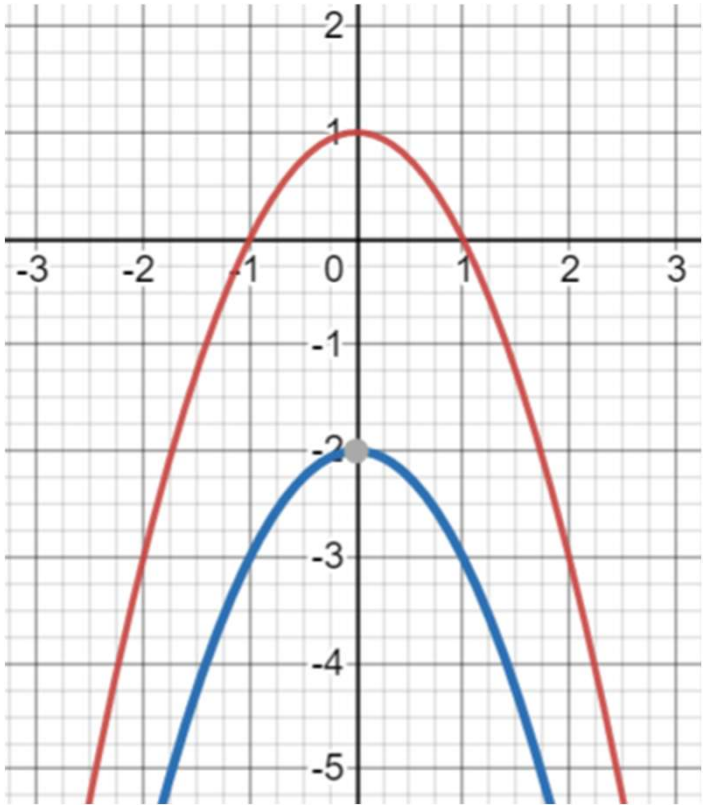
# When the leading coefficient is negative.

Adding a positive constant shifts the function up by that amount.

$y = -x^2 + 1$

Adding a negative constant shifts the function down by that amount.

$y = -x^2 - 2$




$y = -x^2 + 1$

$y = -x^2 - 2$

# Vertex form of a quadratic

Go to Desmos

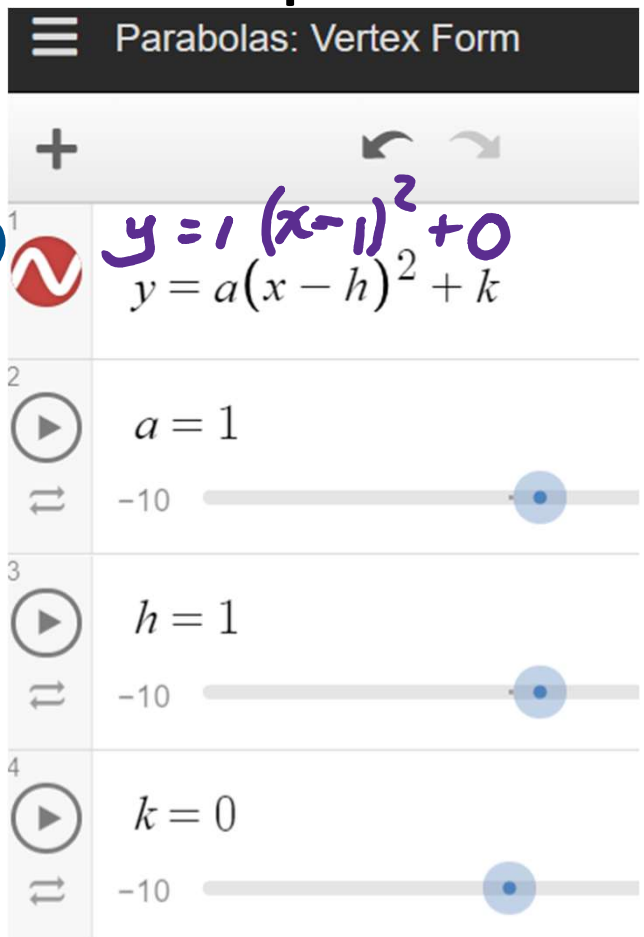
$y = 1(x-0)^2 + 0$

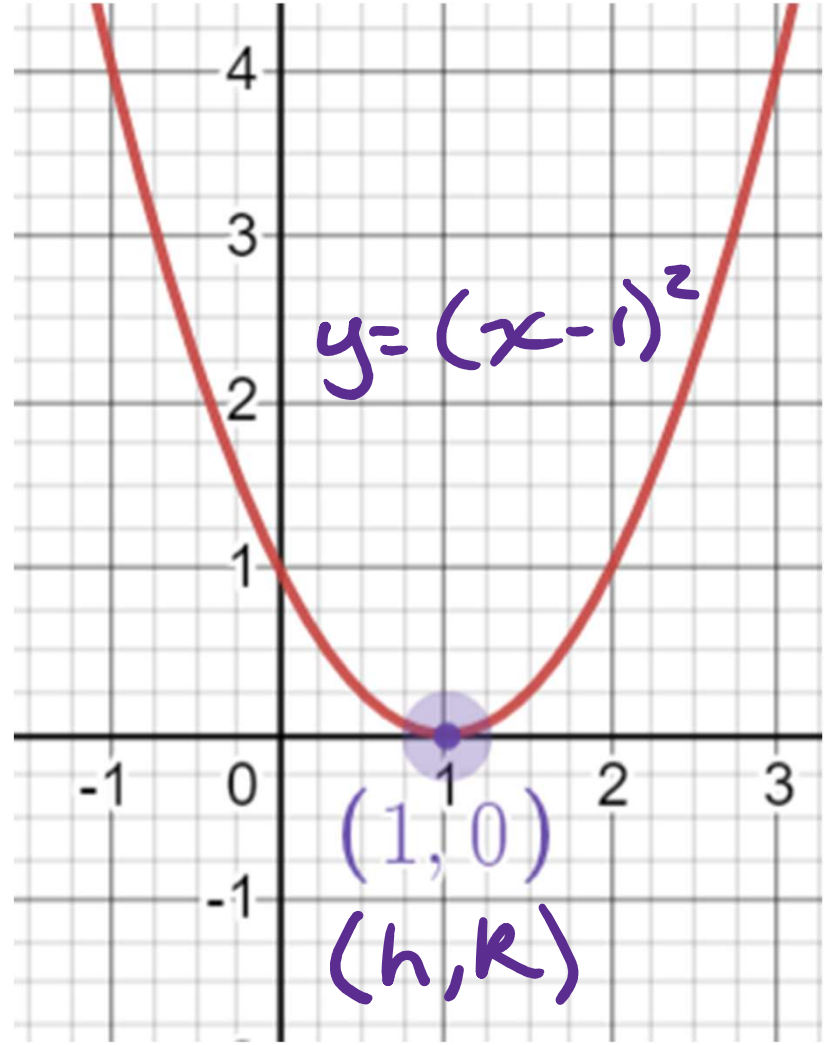
$a = 1$

$h = 0$

$k = 0$

$y = x^2$



Want to see what x is when y is zero. Because the h is inside the function, the x value that makes y zero will be the opposite sign of h.

$$y = (x+h)^2$$

$$0 = (x+h)(x+h)$$

$$x+h = 0$$

$$x = -h$$

$$y = (x-h)^2$$

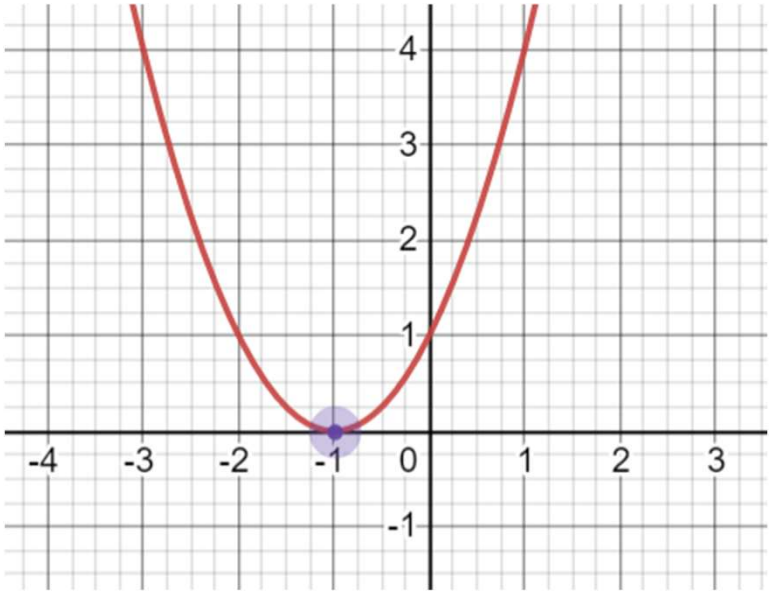
$$0 = (x-h)(x-h)$$

$$x-h = 0$$

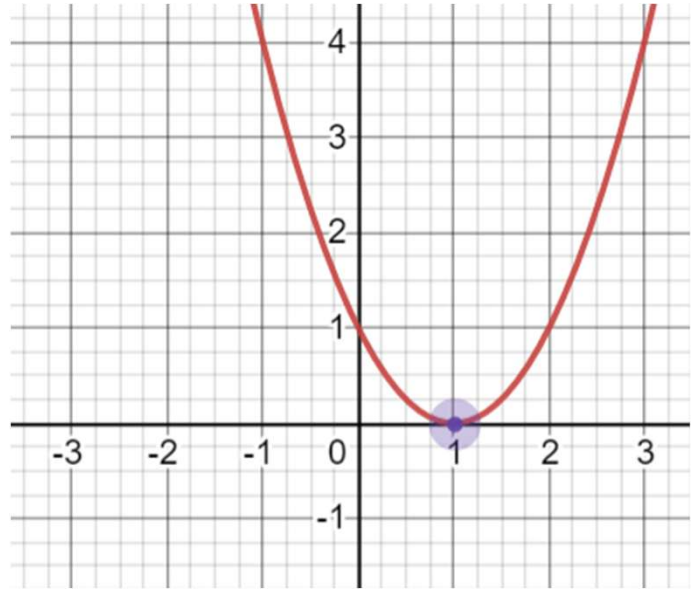
$$x = h$$

Notice that h is an x value and has the same units as x. When the sign inside the function is positive, the parabola is at negative h. When the sign inside the function is negative it is at positive h.

$$y=(x+1)^2$$



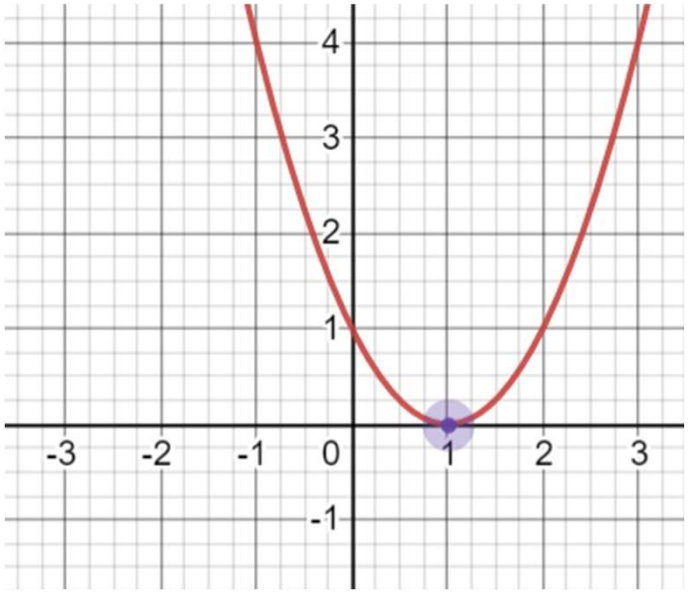

$$y=(x-1)^2$$



The parabola shifts right when it is minus inside the parentheses and h is positive and shifts left when it is plus inside and h is negative.

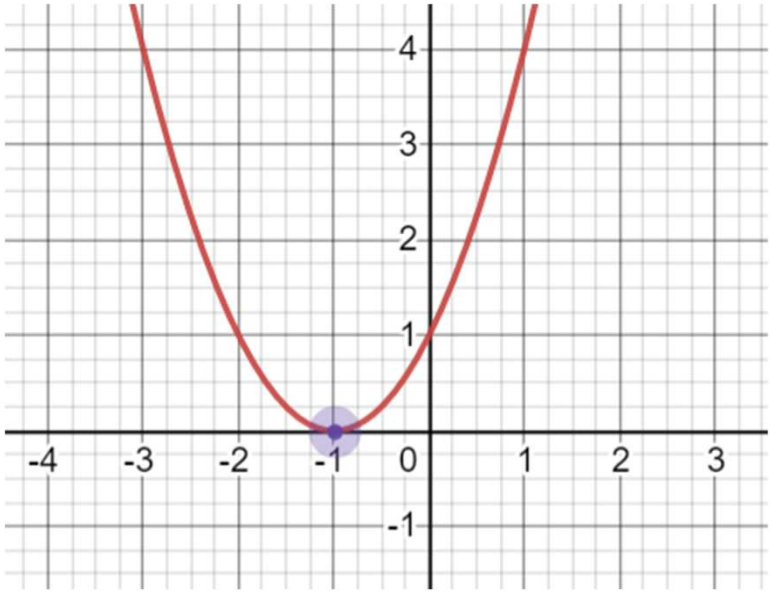
$y = (x - 1)^2$

$y = (x - 1)^2$




$y = (x + 1)^2$

$y = (x + 1)^2$



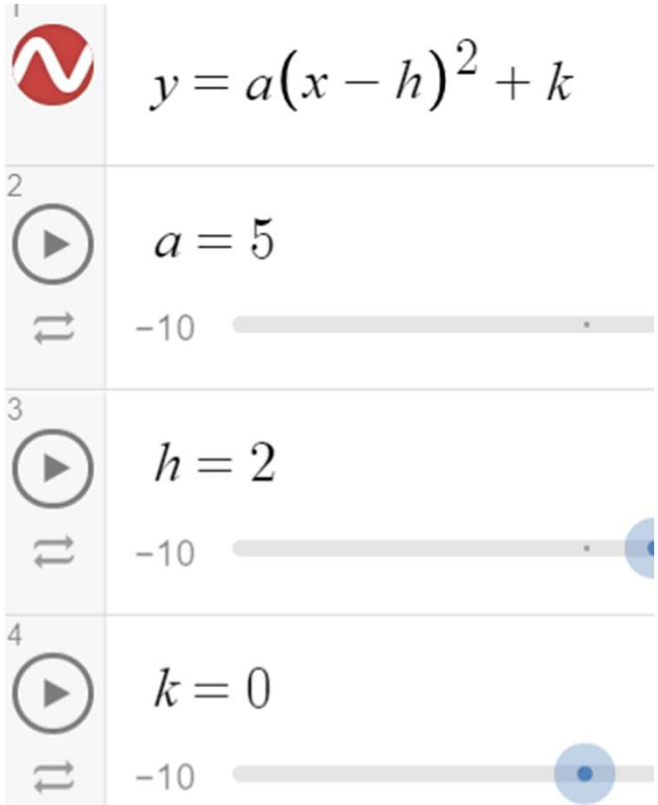
$y = x^2 + 5 \uparrow 5$

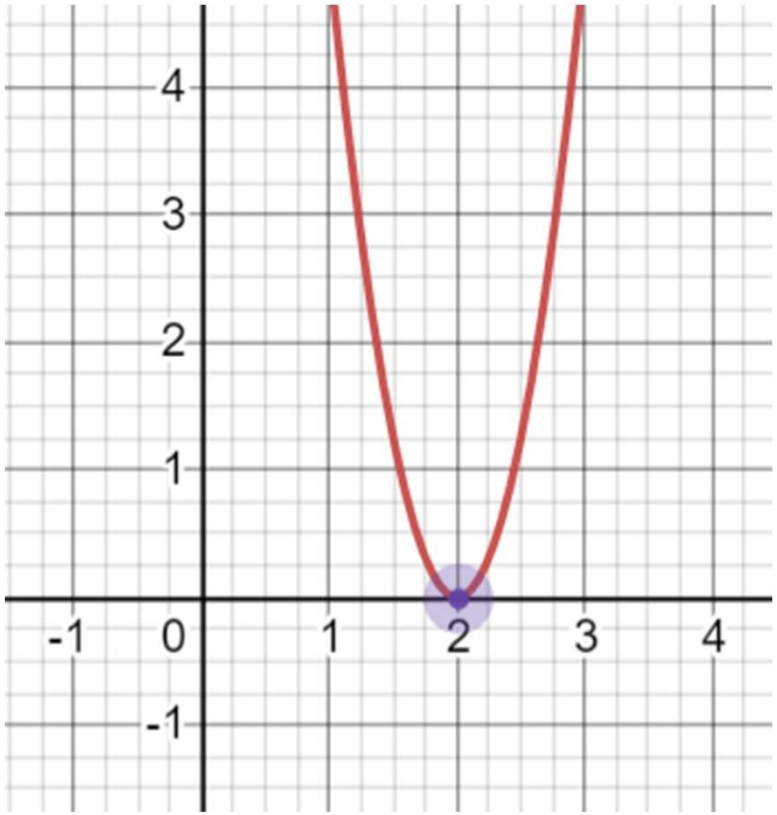
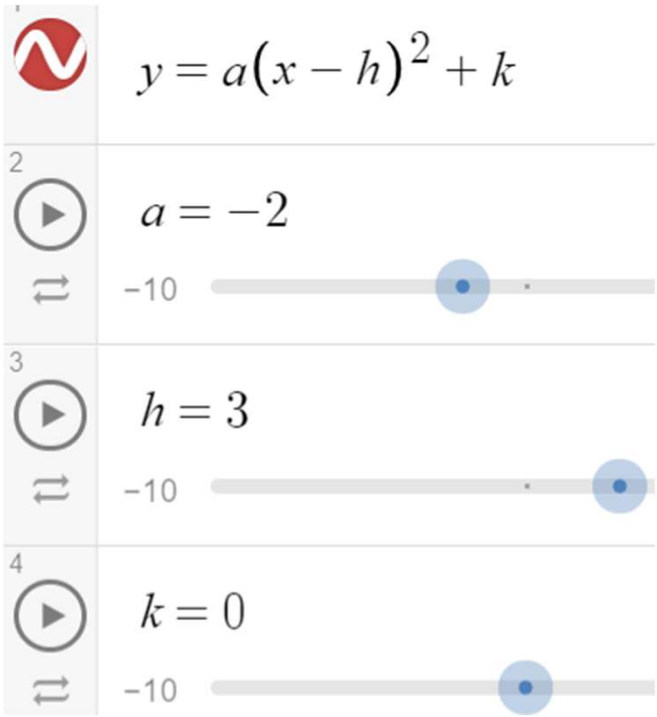
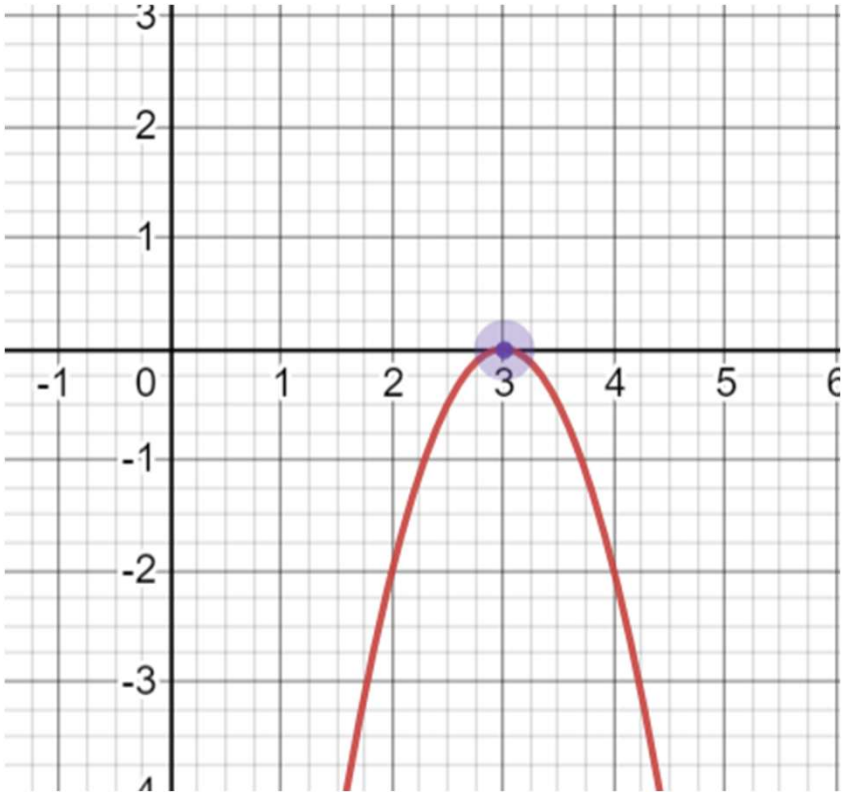
$y = x^2 - 2 \downarrow 2$

$y = (x-1)^2 + 5 \uparrow 5$ R1

$y = (x-2)^2 - 3 \downarrow 3$ R2

The a is how steep or shallow, positive a is smiley or concave up, negative is concave down.






# Here a is negative so concave down.